CloudSync Pro

Enterprise Product Manual

Version: 4.2.0

Chapter 1: Product Overview

CloudSync Pro provides resilient, real-time enterprise data backup and hybrid cloud synchronization.

Designed for distributed engineering and finance operations across multi-cloud environments.

Automatic conflict resolution ensures seamless team collaboration without file corruption.

System Requirements & Pre-requisites

Operating Systems: Windows 10/11 64-bit, macOS 12 Monterey or later, Ubuntu 22.04 LTS.

Memory & CPU: Minimum 8 GB RAM, 4 Core CPU, 10 GB SSD scratch space.

Network Ports: Outbound TCP port 443 must remain unrestricted for TLS handshakes.

Ensure local storage write permissions are granted to the background service daemon.

Chapter 2: Backup Configuration

- The CloudSync Pro platform supports auto-backup scheduling under Settings > Backup.
- Supported storage protocols for automated synchronization are SFTP, S3, and GCS.
- Incremental sync intervals can be set from 5 minutes to 24 hours depending on network bandwidth.
- Bandwidth throttling can be applied during peak operational hours to prioritize internal traffic.

Storage Endpoints and Encryption Standards

- All files in transit are encrypted using TLS 1.3 protocol.
- Data at rest utilizes AES-256 GCM envelope encryption.
- Customer Managed Encryption Keys (CMEK) are supported for enterprise tenants via AWS KMS and HashiCorp Vault.
- Zero-knowledge encryption keys can be enabled by account administrators under Security Settings.

Chapter 3: User Management and Role-Based Access Control

- Admins can assign four default user roles: Organization Owner, Team Admin, Member, and Viewer.
- Custom permission policies can be defined using YAML configuration templates.
- Team directories can be restricted to specific IP subnet ranges.
- Admins can revoke active sessions immediately from the centralized security dashboard.

Single Sign-On (SSO) Integration

- CloudSync Pro integrates seamlessly with Okta, Azure Active Directory, and Google Workspace.
- SAML 2.0 and OpenID Connect (OIDC) identity assertions are supported.
- Directory sync allows automatic provisioning and de-provisioning of users using SCIM protocols.
- Session timeout for web and desktop clients defaults to 8 hours unless modified by policy.

Chapter 4: Account Security & Multi-Factor Authentication (2FA)

To reset administrative 2FA, contact the dedicated internal security team with your official employee verification token.

> **Security Policy Warning:** Never share SMS OTP or authentication codes under any circumstances.

Enforce hardware security keys (FIDO2/WebAuthn) for high-privilege administrator accounts.

Failed login attempts are locked for 30 minutes after five consecutive incorrect attempts.

Audit Logging and Compliance Reporting

- All access logs and file transfer transactions are retained for 365 days by default.
- Audit trails are immutable, cryptographically signed, and exportable via CSV or Syslog stream.
- Compliant with SOC2 Type II, HIPAA, and GDPR data handling requirements.
- Log export tasks can be scheduled to forward events directly to external SIEM systems.

Chapter 5: Performance Tuning and Bandwidth Throttling

- Network bandwidth limits can be configured per device under Preferences > Bandwidth.
- Peak and off-peak rate limiting prevents local bandwidth congestion during critical business hours.
- Chunk multi-threading allows large files exceeding 1 GB to transfer in parallel streams.
- Local caching thresholds can be adjusted to minimize round-trip latency for frequently accessed files.

Troubleshooting Common Connectivity Errors

Error Code	Description and Resolution
E-104	Failed handshake with S3 gateway. Verify bucket permissions and IAM policy.
E-208	Rate limit reached. Increase backoff window in advanced gateway configuration.
E-312	DNS resolution timeout. Switch to static fallback DNS entries in config.json.

Check local firewall logs if background socket reconnection attempts continue to fail.

Chapter 6: API Access and Webhook Integrations

- REST API access tokens can be generated inside the developer settings dashboard.
- Webhook notifications are signed using HMAC-SHA256 headers for signature verification.
- Rate limits for API calls are capped at 100 requests per minute per IP address.
- API payload schema specifications are published and maintained in the OpenAPI documentation.

Chapter 7: Billing, Subscriptions and Cancellation Terms

- Refunds are processed within 5-7 business days for annual tier subscriptions.
- To claim a full refund, annual subscriptions must be cancelled within 30 days of the renewal date.
- Monthly tier subscriptions are billed continuously and are strictly non-refundable.
- Support escalation tier depends directly on the active subscription level (Standard vs Enterprise).